Practice School II Report (Mid semester), 2026

GitHub Issue Milestones Completion and Refactoring for the Talawa Project

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CERTIFICATE

This is to certify that the internship project work entitled **"GitHub Issue Milestones Completion and Refactoring for the Talawa Project"** submitted by **Divyanshu Gautam**(2022Btech034) towards the fulfillment of the requirements for the degree of **Bachelor of Technology in Computer Science and Engineering** of JK Lakshmipat University Jaipur is the record of work carried out by him under my supervision and guidance. In my opinion, the submitted work has reached a level required for being accepted for Practice School-I examination.

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I would like to thank Dr. Satveer Singh(My Faculty Supervisor), Department of CSE at JKLU for giving me opportunity to work at Palisadoes Foundation. Their support and facilitation have been greatly appreciated.

Sincerely yours,

Divyanshu gautam
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ABSTRACT

This project focuses on the comprehensive resolution and implementation of a defined set of **11 specific GitHub issues** within the Talawa project, targeting the talawa-admin (Admin Portal) and talawa-api (Backend API) repositories. The primary goal is to enhance the application's stability, user experience, and overall code quality by addressing existing technical debt and functional gaps.

The scope is divided into two core categories: Admin Refactoring, which involves critical UI/UX improvements, fixing non-functional donation and campaign processes, converting legacy reporting tables to modern DataTable components, and refactoring reusable components; and **Setup & Installation Fixes**, which resolves critical bugs and inconsistencies to ensure a smoother onboarding and development environment setup. Successful completion will be verified through the closure of all associated GitHub issues, contributing to a more robust and scalable Talawa platform.

CONTENTS

CERTIFICATE 2
ACKNOWLEDGEMENTS 3
ABSTRACT 4
CHAPTER 1 – ABOUT THE ORGANIZATION 6
Mission 6
Vision and Goals 6
History 6
CHAPTER 2 – PRODUCT & ENHANCEMENTS 7
Talawa Mobile App 7
Talawa-Admin Portal 7
People Management 7
Events and Posts 7
Plugins and Donations 8
Talawa-API Backend 8
CHAPTER 3 - EVENTS 8
Event Creation Basics 9
Recurring Events 9
Action Items and Agenda 10
Volunteer Management 10
Event Statistics 10
CHAPTER 4 - TOOLS AND TECH USED 11
CHAPTER 5 - ROLES AND RESPONSIBILITIES 12
Log table: 12
CHAPTER 6 - FUTURE PLANS 13
CONCLUSION 14

CHAPTER 1 – ABOUT THE ORGANIZATION

The Palisadoes Foundation is a non-profit organization founded in 2016 by Jamaican diaspora technology professionals based in Cupertino, California. It focuses on promoting STEM education, particularly software engineering, through open-source projects and internships.

Mission

The foundation promotes software engineering education internships for university students in under-served communities, with a demographic emphasis on Black, Brown, Latinx, and Indigenous groups. Its geographic focus spans North America and the Caribbean, aiming to cover costs via revenue from software projects.

Vision and Goals

Its vision is equitable access to life opportunities for STEM education recipients. Long-term goals include fostering global innovators by encouraging open-source participation, starting with programs like the Calico Challenge, which offers stipends to students contributing to open-source under mentors.

History

Created to advance technology in Jamaica, it expanded to underserved North American communities. Key initiatives include the Calico Challenge, partnering with groups like the Jamaica Computer Society.

CHAPTER 2 – PRODUCT & ENHANCEMENTS

Talawa is an open-source community management platform featuring a mobile app, admin portal, and API backend. Its components—Talawa mobile app, Talawa-admin web portal, and Talawa-API—enable member engagement, event management, and administrative control through social features and secure data handling.

Talawa Mobile App

Talawa provides social media-like features for community interaction in a mobile app. Users can share posts, create group events, chat with members, view feeds, receive notifications, and manage reminders to strengthen community ties.

- Supports donations (recurring or one-time) visible in admin contributions.
- Tracks volunteer time and activity tasks.
- Customizable with plugins installed via admin portal.docs-admin.

Talawa-Admin Portal

Talawa-admin is a web-based dashboard for organization leaders to manage communities efficiently. Admins oversee memberships, events, posts, donations, and plugins from an intuitive interface.

People Management

Admins approve/reject membership requests, add/remove members, set permissions, elevate to admin, and ban users with reasons. Super admins have extra powers to manage other admins and organization.

Events and Posts

Create/delete events, view active ones, moderate posts (delete, archive, hide likes, manage comments/sharing), and create group chats. Track attendance and volunteer engagement.

Plugins and Donations

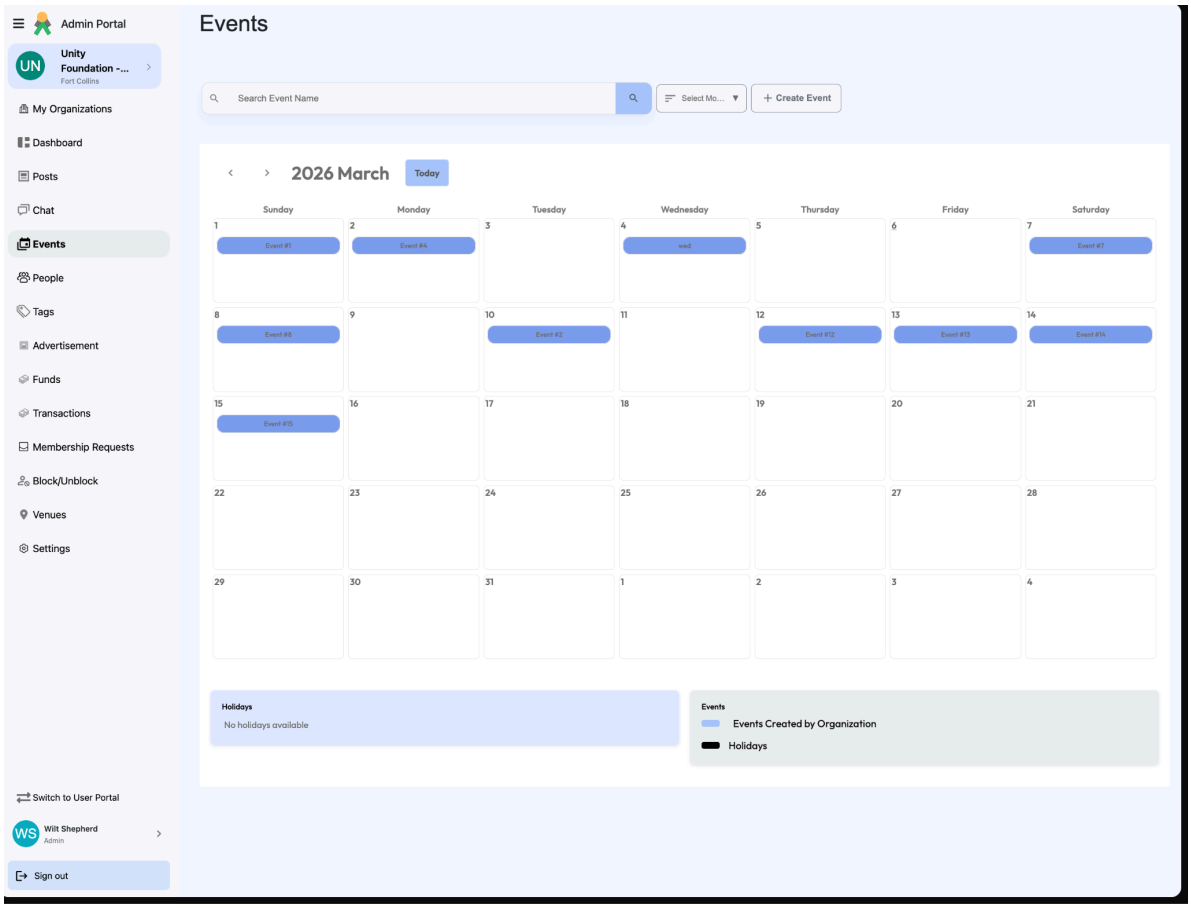
Access a plugin store to install/uninstall features for the mobile app. Monitor contributions and donations from app users..

Talawa-API Backend

Talawa-API serves as the central backend using GraphQL, Node.js, Express, and MongoDB for secure data access between app and admin. It centralizes functionality as the source of truth, supports multi-tenancy, pagination, and internationalization (English, Hindi, Chinese, Spanish, French).

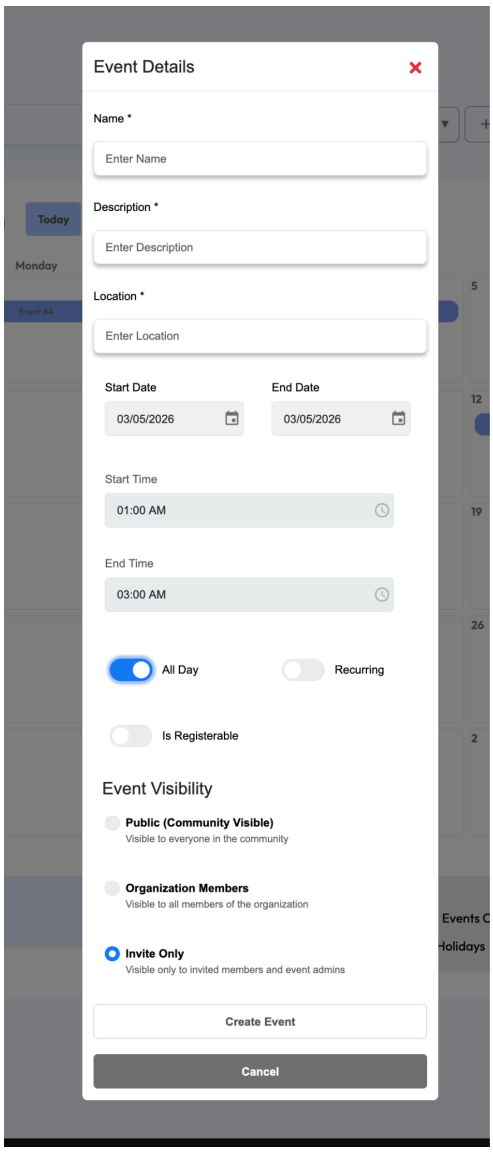
CHAPTER 3 - EVENTS

Talawa's event management system streamlines planning, coordination, and tracking for community organizations through its admin portal and mobile app. Key features include calendars, volunteer assignments, action items, and analytics, all integrated with group chats and notifications



Event Creation Basics

Admins and authorized users create events via the admin dashboard or mobile app, specifying title, description, date, time, location, and recurrence options. Events auto-generate group chats for invitees to facilitate real-time coordination



Recurring Events

Events can be set as recurring (e.g., weekly on Tuesdays/Thursdays), mimicking Google Calendar functionality for meetings, workshops, or support groups. Admins define frequency, end dates, and exceptions directly in the creation form

Action Items and Agenda

Each event has a dedicated dashboard with an "Event Actions" tab for listing roles, responsibilities, and tasks assigned to volunteers or participants. Organizers add, edit, or check off items in real-time via mobile or tablet, ensuring agendas stay on track.

Volunteer Management

Volunteers are assigned specific roles per event, with tracking for time logged and tasks completed. Admins monitor participation to build reports on engagement trends.

Event Statistics

Dashboards provide analytics like attendee counts (who's coming, RSVPs), attendance trends, volunteer activity, and overall participation metrics. View all events by day or filter for insights into recurring patterns

Feature	Description
Attendance Tracking	Real-time RSVPs and check-ins with reports on frequency.
Integrations	Links to posts, donations, and notifications; supports venue booking (name, capacity, images).
Moderation	Admins edit/delete events, manage invites, and broadcast updates. YouTube
Mobile Sync	Users view calendars, join chats, and receive reminders in the app.

CHAPTER 4 - TOOLS AND TECH USED

Talawa platform employs a modern, open-source tech stack for robust community management across its mobile app, admin portal, and API backend.

Node.js:
Node.js is a JavaScript runtime used to build scalable server-side applications. It handles asynchronous operations efficiently and powers the backend logic of the application.

Express:
Express is a minimal web framework for Node.js used to create APIs and manage routing and middleware. It simplifies handling HTTP requests and structuring backend services.

GraphQL (Apollo Server):
GraphQL is a query language that allows clients to request only the data they need. Apollo Server helps implement GraphQL APIs and manage queries and mutations efficiently.

React:
React is a JavaScript library for building dynamic and component-based user interfaces. It is used to develop the Talawa-admin portal frontend.

PostgreSQL:
PostgreSQL is a relational database used to store and manage structured application data reliably.

GitHub:
GitHub is used for version control and collaboration, allowing developers to manage code changes and work together through repositories.

Docker:
Docker is used to containerize the application, ensuring consistent environments across development and deployment.

TypeScript:
TypeScript adds static typing to JavaScript, improving code reliability and maintainability across the project.

CHAPTER 5 - ROLES AND RESPONSIBILITIES

- **Software Development:** Contribute to the development of open-source software projects by implementing new features, fixing bugs, and improving existing functionality.
- **Project Maintenance:** Maintain assigned repositories by reviewing code, managing issues, updating dependencies, and ensuring the stability and performance of the project.
- **Code Review & Quality Assurance:** Follow coding standards and best practices, participate in code reviews, and ensure code quality and maintainability.
- **Mentoring Contributors:** Guide new contributors in understanding the project structure, development workflow, and open-source collaboration practices.
- **Collaboration:** Work with other developers and maintainers through GitHub to coordinate development, manage pull requests, and resolve technical issues.
- **Documentation:** Assist in maintaining clear technical documentation for features, APIs, and project setup to support contributors and users.
- **Learning & Skill Development:** Enhance technical, collaborative, and leadership skills through active participation in open-source development activities.

Log table:

Name	Link
Feature Request: Update sample database to include more recurring events (Issue #4312)	https://github.com/PalisadoesFoundation/talawa-api/issues/4312
Test event generation windows (PR #4897)	https://github.com/PalisadoesFoundation/talawa-api/pull/4897
Created test file for src/drizzle/tables/eventAttendees.ts (PR #4898)	https://github.com/PalisadoesFoundation/talawa-api/pull/4898
test: add comprehensive tests for leakyBucket utility (PR #5108)	https://github.com/PalisadoesFoundation/talawa-api/pull/5108
test: add unit tests for authorization utility (PR #5134)	https://github.com/PalisadoesFoundation/talawa-api/pull/5134
added test coverage for file src/drizzle/tables/tagFolders.ts (PR #4953)	https://github.com/PalisadoesFoundation/talawa-api/pull/4953

test: improve Organizations test coverage to be more than 96% (PR #7172)	https://github.com/PalisadoesFoundation/talawa-admin/pull/7172
Refactor Event components to use ProfileAvatarDisplay (PR #7095)	https://github.com/PalisadoesFoundation/talawa-admin/pull/7095
test: Add unit tests for User.state resolver (PR #5189)	https://github.com/PalisadoesFoundation/talawa-api/pull/5189

CHAPTER 6 - FUTURE PLANS

Now (ongoing):

- Actively working on **GitHub open issues** across Talawa Admin + API
- **Reviewing PRs** and helping ensure quality/consistency
- **Mentoring newcomers** (onboarding, guidance, code-review support)
- **Current focus:** Primarily contributing to the **backend (Talawa API)**

Issues I'm currently tracking/working on:

- Populate recurring events in the sample DB
- Setup2Fix - Tracker
- Install2Fix - Tracker
- Install2Fix - Tracker
- Error in manage Tags - Admin Portal
- Reusable Components Refactoring - Parent Issue
- TableFix - Master Plan
- Donations don't work
- Standardize page headers
- Campaigns & pledges
- Campaigns & pledges

CONCLUSION

The mid-semester period for the **GitHub Issue Milestones Completion and Refactoring for the Talawa Project** has established a strong foundation for enhancing the Talawa platform's stability and maintainability. The core objective of comprehensively resolving and implementing a defined set of **11 specific GitHub issues** across the talawa-admin (Admin Portal) and talawa-api (Backend API) repositories is well underway.

Key progress includes addressing both **Admin Refactoring**—such as critical UI/UX improvements, refactoring reusable components, and fixing non-functional donation and campaign processes—and crucial **Setup & Installation Fixes** to streamline the onboarding and development environment. The work directly contributes to technical debt reduction and an improved overall user experience.

Moving forward, the focus remains on the active resolution of the remaining open issues, particularly within the **Talawa API backend**. This ongoing effort, combined with continued participation in **PR reviews** and **mentoring newcomers**, ensures a commitment not only to feature delivery but also to the long-term quality and scalability of the open-source Talawa platform. Successful completion will be verified by the final closure of all associated GitHub issues, cementing a more robust community management solution.